

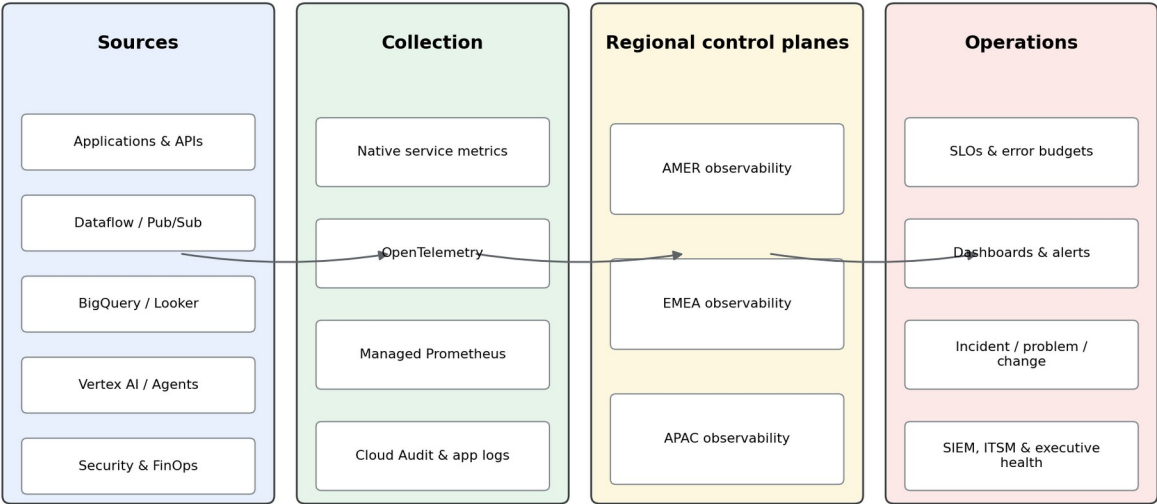
Enterprise Google Cloud Data Platform Blueprint

VOLUME 11

Monitoring, SRE and Operations

Enterprise Reference Architecture and Implementation Blueprint

Enterprise Observability Architecture



Legend: solid arrows show primary telemetry, control or accountability flow. Regional and trust boundaries must be validated per workload.

Enterprise observability, reliability and operations control model

Architecture Baseline 1.0 | 12 July 2026 | Google Cloud

Classification: Enterprise architecture reference. Validate current service capabilities, quotas, locations, retention, pricing, support and regulatory requirements before implementation.

Document Control

Field	Value
Document	Volume 11 - Monitoring, SRE and Operations
Series	Enterprise Google Cloud Data Platform Blueprint
Version	1.0
Status	Detailed reference architecture for adaptation and implementation
Date	12 July 2026
Owner	Enterprise Site Reliability Engineering and Operations Architecture
Approvers	CIO, CDO, CISO, Enterprise Architecture, Platform Engineering, Domain Product Owners, SRE, Security Operations, FinOps, Business Continuity and Service Management
Audience	Executives, service owners, platform/domain engineers, SRE, operations, security, data governance, FinOps, audit and vendor management
Review cycle	Quarterly and after any material incident, service/product change, regulatory change or repeated SLO failure
Primary dependencies	Volumes 01-10; Volume 12 DR and Business Continuity; Volume 13 Terraform. Volume 09 DevSecOps is an explicit implementation dependency and is not assumed complete.
Scope	Operations strategy, SRE operating model, ownership, SLIs/SLOs/SLAs, error budgets, observability, service metrics, dashboards, alerting, on-call, incident/problem/change/capacity management, readiness, resilience and support.

Guidance classification. Mandatory controls apply to Tier 0, Tier 1, Restricted and Regulated workloads unless a time-bound exception is approved. Recommended patterns are enterprise defaults. Conditional patterns require workload evidence and current product validation.

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1. Executive Summary

Volume 11 defines how the enterprise data and AI platform is operated as a portfolio of business services rather than as a collection of independent cloud resources. It establishes a federated SRE model in which central platform SRE provides common observability, incident-command and reliability governance, while platform and domain teams own their service outcomes, SLOs, runbooks and on-call response.

Executive decision: reliability is measured at the consumer boundary - freshness, quality, latency, successful analytics and AI outcomes - and is governed through SLOs and error budgets. Raw infrastructure thresholds support diagnosis but do not define service health.

Outcome	Mandatory target
Measurable service health	Every Tier 0/1 service has approved owner, critical user journeys, SLIs, SLOs, dependencies and error budget.
Actionable detection	Every page-level alert represents likely user impact or rapid SLO burn and has an exercised runbook.
Federated accountability	Central SRE owns the platform and standards; platform/domain teams own their services and consumer communication.
Regional control	Telemetry collection, storage and aggregation align with AMER, EMEA and APAC residency and access boundaries.
Operational evidence	ORR/PRR, incident response, PIR, capacity and resilience decisions are supported by reproducible evidence.
Cost and security	Telemetry is governed data with classification, retention, access, cardinality and cost controls.

- Tier 0 production objective: 99.9% to 99.99% service availability as approved, RTO < 1 hour and RPO < 5 minutes, subject to service-specific constraints.
- Tier 1 objective: RTO < 4 hours and RPO < 15 minutes; Tier 2: RTO/RPO < 24 hours.
- A telemetry blind spot is itself a high-severity reliability event for Tier 0 and Tier 1 services.

2. Inherited Context, Scope and Assumptions

- The enterprise is a global Fortune 100 organization with 10-50 PB of historical data, 50-250 TB daily growth, billions of events, hundreds of systems, thousands of pipelines and thousands of users.
- The architecture uses AMER, EMEA and APAC regional data planes, centrally governed platform capabilities and federated domain-owned data products.
- BigQuery is the primary analytical engine; Cloud Storage/BigLake provide durable and open-format data layers; Pub/Sub, Dataflow, Datastream, Dataform and Managed Airflow implement ingestion and transformation.
- Looker provides governed semantic analytics; Vertex AI and Gemini support ML, GenAI, RAG and agent workloads.
- Identity is group-based and least-privilege. Private-by-default networking, VPC Service Controls, selective CMEK, centralized audit, Terraform authority and evidence-based production gates are inherited controls.
- Security classifications are Public, Internal, Confidential, Restricted and Regulated. Telemetry inherits the highest applicable classification when it contains sensitive identifiers, prompts, queries, schemas or payload fragments.
- Volume 10 defines cost allocation and FinOps. Volume 12 defines recovery. This volume connects both through operational metrics, capacity, error budgets and incident evidence.
- Volume 9 DevSecOps has not yet been generated in this sequence. Observability-as-code and release-gate designs are defined here but remain implementation dependencies on Volume 9 and Volume 13.

Item	Classification	Volume 11 treatment
Confirmed	Master prompt scope, inherited platform services and regional data-plane model	Used as baseline.
Architecture assumption	Regional observability projects and controlled enterprise views	Default pattern pending workload/location validation.
Mandatory control	Tier 0/1 SLOs, on-call, runbooks, tested alerts, incident and PIR process	Required before production approval.
Recommended control	OpenTelemetry and managed Prometheus standards	Default for portable application metrics/traces.
Conditional capability	App Hub/Application Monitoring and advanced automated remediation	Adopt after product/location/maturity validation.
Open decision	Enterprise ITSM/SIEM/paging product integration and support-contract tiers	Resolve during implementation.

3. Operations Strategy

The operating strategy applies five principles:

- Business-service first: observe and operate the end-to-end consumer journey, not only individual Google Cloud services.
- Symptoms before causes: page on freshness, quality, availability, latency and error-budget burn; use component signals for diagnosis.
- Federated ownership: ownership follows the service/data product; shared platforms provide standard capabilities and incident command.
- Automation with guardrails: automate collection, deployment, evidence and bounded remediation; gate destructive or wide-blast-radius actions.
- Continuous learning: error budgets, PIRs, recurring-problem analysis, capacity reviews and resilience exercises drive prioritized improvement.

Operations capability	Implementation pattern	Evidence
Service registry	Service ID, tier, owner, region, data classification, dependencies, SLOs and support route	Versioned service record
Telemetry contract	Required resource attributes, metric/log/trace standards and forbidden sensitive fields	Validated schema and CI result
Reliability control	SLOs, error budgets and change policy	SLO configuration and monthly report
Incident response	Severity, command roles, communication cadence and runbooks	Incident timeline and PIR
Capacity/performance	Demand, quota, reservation, headroom and saturation reviews	Capacity model and actions
Assurance	ORR/PRR and resilience tests	Signed evidence package

4. SRE Operating Model

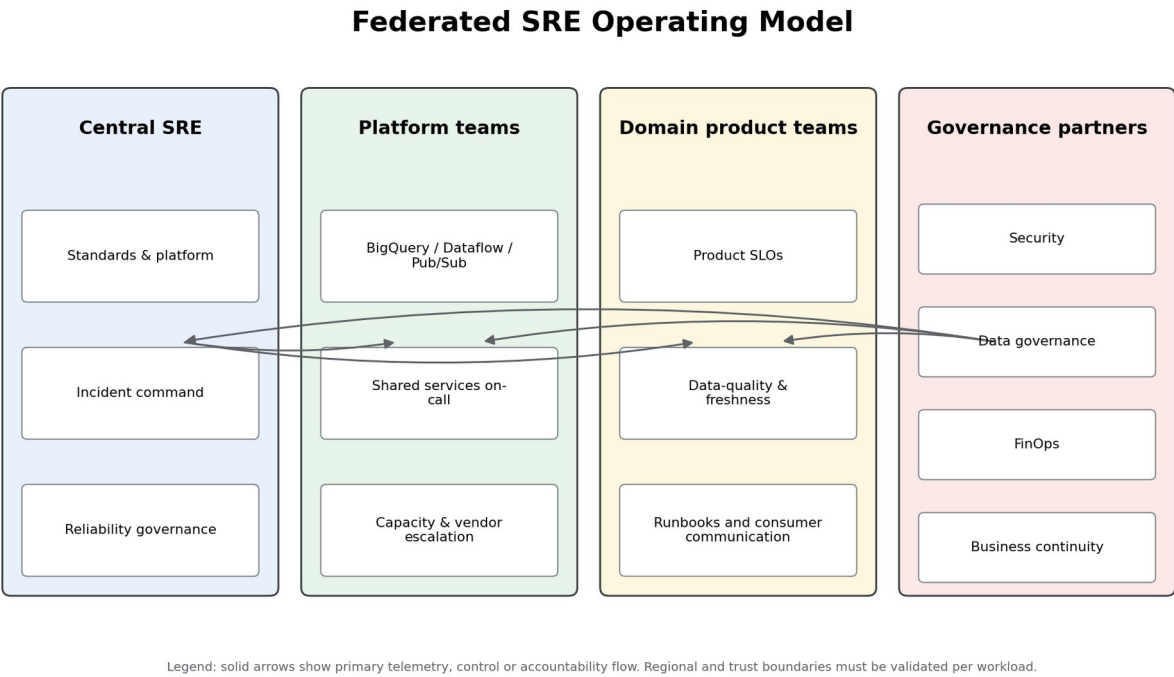


Figure 1 - Federated SRE operating model

Central SRE is a platform and governance function, not the default owner of every application or data product. It operates the regional observability foundation, defines reliability standards, leads major incidents, measures portfolio health and coaches product teams. Platform teams operate shared services. Domain teams own business semantics, freshness, quality, consumer impact and data-product response.

Role	Accountability	Minimum operational duty
Business service owner	Business impact, target service level and risk acceptance	Approve SLO, error-budget policy and major incident communication.
Platform SRE	Common platform reliability and incident command	Operate observability, on-call standards, major incident process and reliability reviews.
Platform owner	BigQuery/Dataflow/Pub/Sub/AI/analytics shared capability	Capacity, service alerts, vendor escalation and platform runbooks.
Domain product owner	Data-product reliability and consumer outcomes	Freshness/quality SLO, product on-call, reconciliation and consumer communication.
Security operations	Security monitoring and containment	Operate SIEM/SOAR, evidence preservation and security incident escalation.
FinOps	Cost anomaly and capacity economics	Monitor cost/unit economics; support commitment and capacity decisions.
Business continuity	Recovery readiness	Coordinate exercises and recovery governance with Volume 12.

5. Service, Platform and Data-product Ownership

Mandatory: no production service, pipeline, data product, dashboard, model endpoint or alert may be ownerless. Ownership must resolve to a group and an accountable executive/business service owner, not an individual-only mailbox.

Asset	Accountable owner	Responsible operator	Reliability boundary
Cloud observability foundation	Enterprise Platform SRE	Observability platform team	Collection, storage, alerting and notification availability
BigQuery shared platform	Analytics/Data Platform Owner	BigQuery platform on-call	Job success, reservation health, access and core platform dependencies
Dataflow/Pub/Sub shared platform	Streaming Platform Owner	Streaming platform on-call	Job runtime, backlog, quotas and templates
Domain data product	Domain Product Owner	Domain data engineering/on-call	Freshness, quality, schema contract, consumer impact
Looker semantic service	Analytics Product Owner	BI/analytics on-call	Explore/dashboard availability, semantic correctness and query experience
Vertex AI service	AI Product Owner	AI platform/product on-call	Endpoint availability, latency, model version and safety/evaluation controls
Security telemetry	CISO delegate	Security operations	Detection delivery, audit evidence and response
FinOps telemetry	CFO/CIO delegate	FinOps	Allocation completeness, anomalies and unit cost

Operational ownership must be synchronized with IAM groups, service catalog/CMDB, repositories, Terraform labels, dashboards, alert policies, runbooks, incident tooling and cost allocation. A quarterly drift check identifies mismatches and blocks production promotion for critical services.

6. Service Tiering and Critical User Journeys

Tier	Business impact	Availability/SLO posture	Operational coverage
Tier 0	Material enterprise, regulatory, safety, financial or customer impact	Typically 99.95%-99.99%; RTO <1h, RPO <5m where technically supported	24x7 on-call, rapid burn alerts, tested failover/recovery, quarterly exercises
Tier 1	Major business process or high-value analytics/data product	Typically >=99.9%; RTO <4h, RPO <15m	24x7 or extended coverage, product runbooks, semiannual resilience test
Tier 2	Important but delay-tolerant service	Business-hour or approved target; RTO/RPO <24h	Business-hour response, ticketing and annual test
Tier 3	Development, sandbox or noncritical capability	Best effort with budget/cost guardrails	No page; owner ticket and lifecycle controls

Critical user journeys are end-to-end outcomes. Examples include source event to certified data product, executive query to approved KPI, regulated report production, AI request to grounded response, and high-severity finding to SIEM case. Each journey has an owner, dependency map, instrumentation and approved exclusions.

Anti-pattern: assigning Tier 0 to every component. Tiering applies to the business service; components inherit operational priority only when their failure threatens the service objective.

7. SLIs, SLOs, SLAs and Error Budgets

Term	Enterprise definition	Use
SLI	Measured ratio, distribution or time-based indicator of service behavior	Quantifies availability, latency, freshness, quality, correctness or telemetry health.
SLO	Internal reliability target over a defined window	Balances reliability with delivery speed and drives alerts/change policy.
SLA	External or contractual commitment with remedies or consequences	Approved by legal/commercial authorities; normally less aggressive than internal SLO.
Error budget	Permitted unreliability: 1 minus SLO, expressed over the window	Governs release risk, reliability investment and exception approval.

Service type	Primary SLI	Important companion indicators
Streaming data product	p99 event-time-to-certified latency and successful record ratio	Pub/Sub backlog, Dataflow system lag, DLQ, duplicates, reconciliation
Batch data product	Completion before cutoff, freshness and critical quality pass ratio	DAG duration, task retries, source completeness
BigQuery analytics	Successful valid queries and p95/p99 response experience	Queued time, slots, bytes processed, reservation saturation
Looker/BI	Dashboard/explore success and time-to-first-result	Cache behavior, query errors, underlying BigQuery health
Vertex AI endpoint	Success rate and p95/p99 latency	Endpoint traffic, compute, model version, drift/evaluation and dependency health
Observability foundation	Critical telemetry delivered and page notifications acknowledged	Collector drops, sink errors, canary age, notification tests

Error-budget policy: at 50% budget consumed early in the window, review reliability risk; at 75%, require owner action; at 100%, freeze nonessential risky change until the budget recovers or accountable exception is approved.

7.1 Representative SLI and SLO Catalog

ID	Service/Product	SLI Category	Measurement	Target	Window	Owner	Alert/Action	Notes
SLO-001	Tier 0 data-product freshness	Freshness	p99 age of latest certified partition	99.95% <= 5 min	30-day rolling	Domain Product Owner	Page at 2x/1h or 6x/6h burn	Excludes approved source outage
SLO-002	Tier 1 batch completion	Timeliness	Successful critical DAG runs before business cutoff	99.9%	30-day rolling	Data Engineering	Page if cutoff at risk; ticket on isolated failure	Runbook includes source reconciliation
SLO-003	Streaming event processing	Latency	p99 event-time-to-curated latency	99.9% <= 120 sec	28-day rolling	Streaming Platform	Page on sustained backlog/lag burn	Measured at data-product boundary
SLO-004	BigQuery interactive analytics	Availability	Successful interactive query jobs	99.9%	30-day rolling	Analytics Platform	Ticket on regional or reservation degradation	Excludes invalid user SQL
SLO-005	Vertex AI online prediction	Availability/latency	Successful requests and p95 latency	99.9%; p95 <= approved target	30-day rolling	AI Platform	Page on availability or latency burn	Per endpoint risk tier
SLO-006	Observability pipeline	Telemetry health	Critical metrics/logs/traces delivered within 5 min	99.95%	30-day rolling	Observability Platform	Page on telemetry blind spot	Synthetic canaries required
SLO-007	Security detection delivery	Detection timeliness	High-severity findings delivered to SIEM	99.95% <= 5 min	30-day rolling	Security Operations	Page on sink or routing failure	Separate from incident response SLA
SLO-008	Cost allocation completeness	FinOps quality	Daily spend allocated to owner/product/cost-center	99.5%	Calendar month	FinOps	Ticket when unknown allocation >0.5%	Based on billing export

The complete editable catalog is included in `matrices/sli_slo_catalog.csv`. Production SLOs must define measurement query, window, exclusions, owner, alerting behavior, maintenance policy and consumer impact.

8. Enterprise Observability Architecture

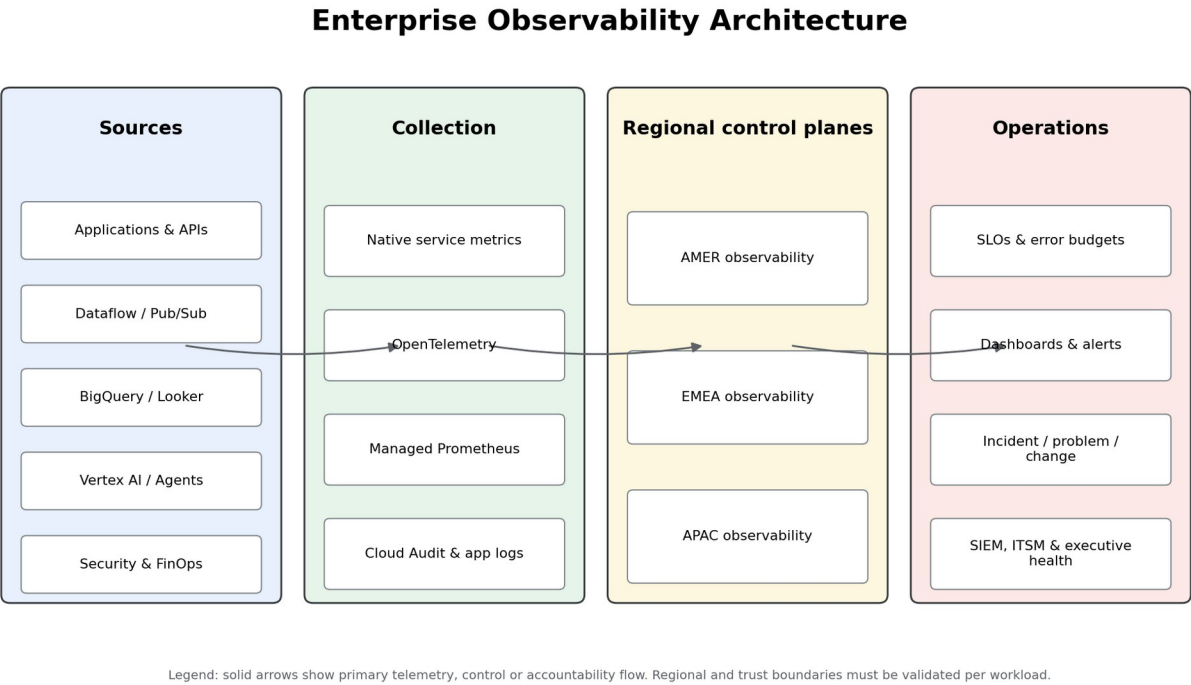
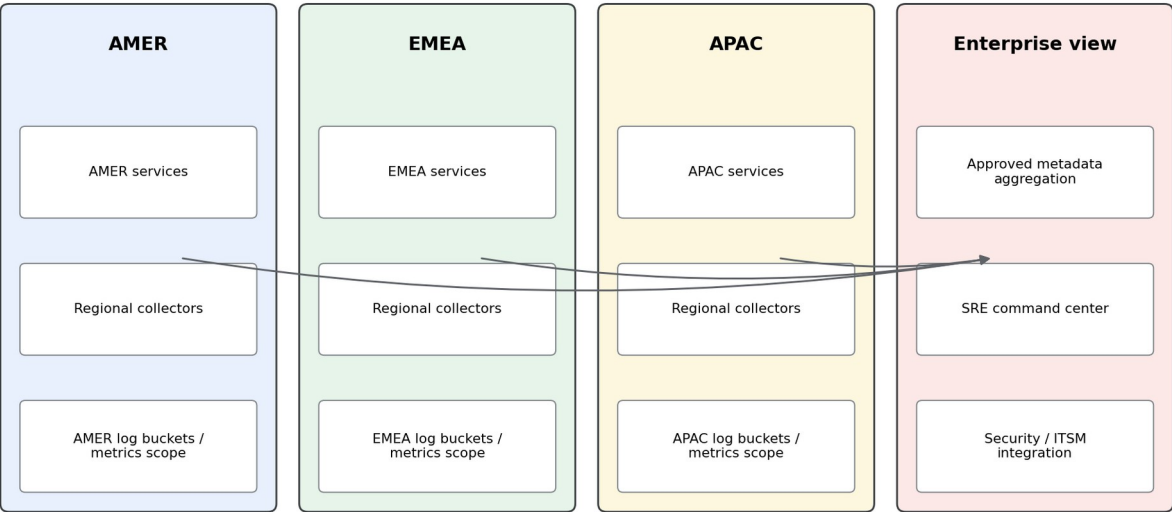


Figure 2 - Enterprise observability architecture

Plane	Capabilities	Primary controls
Instrumentation	Native service metrics/logs; OpenTelemetry; Prometheus format; structured application events	Telemetry contract, service identity, sensitive-field filtering, label budget
Collection	Regional collectors, managed collectors, service integrations and audit log routing	Private/controlled egress, retries, queues, canaries, least privilege
Storage/control	Cloud Monitoring metrics scopes, regional Cloud Logging buckets, trace/error backends	Residency, retention, IAM, VPC Service Controls validation, cost controls
Reliability	SLIs/SLOs, error budgets, dashboards, alerts, uptime and synthetic checks	Ownership, runbook links, tested notification paths, versioned configuration
Operations	On-call, incident command, problem/change/capacity, resilience and vendor support	Severity model, evidence, PIR, approval and continuous improvement
Consumers	SRE, service owners, security operations, FinOps, executives and audit	Purpose-based views and separation of duties

9. Regional Telemetry, Metrics Scopes and Residency

Regional Telemetry, Residency and Aggregation



Legend: solid arrows show primary telemetry, control or accountability flow. Regional and trust boundaries must be validated per workload.

Figure 3 - Regional telemetry topology and controlled enterprise aggregation

Each geography uses a designated observability project and regional logging location. Cloud Monitoring metrics scopes provide multi-project visibility under a scoping project. Enterprise aggregation includes only approved operational metadata and avoids copying sensitive log payloads across boundaries unless legal, privacy and security approvals exist.

Data plane	Default observability pattern	Enterprise aggregation
AMER	AMER scoping project, regional log buckets, regional collectors and notification channels	Portfolio SLO status, incident state, selected capacity and cost KPIs
EMEA	EMEA scoping project and EMEA-approved log location, access groups and collectors	Approved metadata only; no regulated payload export without control mapping
APAC	APAC scoping project, regional log buckets and collectors	Approved metadata only, with country-specific validation where required
Global/shared services	Dedicated global/shared control project where a service is inherently global and approved	Service health and dependency signals linked to regional consumers

Implementation dependency: validate current Monitoring/Logging data-location characteristics, metrics-scope quotas, supported regions and perimeter behavior during detailed design. The architecture does not assume that every telemetry type can be pinned identically.

10. Cloud Monitoring

Cloud Monitoring is the primary managed metrics, dashboard, service-monitoring and alerting control plane. Regional scoping projects provide controlled visibility into platform and domain projects. Policies, dashboards, SLOs and notification metadata are maintained as code and carry owner, service, environment, region, severity and runbook labels.

- Use native service metrics before creating duplicates.
- Create custom metrics only for business-service outcomes such as freshness, quality, certification, reconciliation and incident state.
- Prefer distributions for latency; ratios for availability/quality; time-since-last-good for freshness.
- Monitor the observability system itself through synthetic canaries, policy inventory, API errors and notification tests.
- Do not page on CPU, slots or queue depth alone unless it is a proven leading indicator of imminent user impact.

Representative Terraform alert policy. Full source includes prerequisites, runbook metadata, controlled auto-close and lifecycle protection.

```
# Purpose: SLO-aligned Pub/Sub backlog alert example.
# Prerequisites: notification channel and project APIs are provisioned.
resource "google_monitoring_alert_policy" "pubsub_backlog" {
  project      = var.observability_project_id
  display_name = "prod-streaming-pubsub-oldest-unacked-age"
  combiner     = "AND"

  documentation {
    content = "Runbook: ${var.runbook_base_url}/STR-002"
    mime_type = "text/markdown"
  }

  conditions {
    display_name = "Oldest unacked message age exceeds product threshold"
    condition_threshold {
      filter = "resource.type=\"pubsub_subscription\" AND
metric.type=\"pubsub.googleapis.com/subscription/oldest_unacked_message_age\" AND
resource.label.subscription_id=monitoring.regex.full_match(\"prod-.*\")"
      comparison = "COMPARISON_GT"
      threshold_value = var.oldest_unacked_seconds
      duration = "300s"

      aggregations {
        alignment_period = "60s"
        per_series_aligner = "ALIGN_MAX"
        cross_series_reducer = "REDUCE_MAX"
        group_by_fields = ["resource.label.subscription_id"]
      }
    }

    trigger { count = 1 }
  }
}

notification_channels = [var.primary_oncall_channel]
user_labels = {
  severity = "high"
  service = "streaming-platform"
  runbook_id = "str-002"
  owner = "data-platform-sre"
}

alert_strategy {
  auto_close = "1800s"
  notification_rate_limit { period = "300s" }
}
```

11. Cloud Logging

Cloud Logging provides the enterprise operational and audit event pipeline. Organization and folder sinks route production telemetry to regional buckets and approved security, analytics or archive destinations. Routing is designed before workload onboarding because sink behavior is not a retroactive recovery mechanism for logs that were never routed.

Log class	Default retention/route principle	Access
Administrative and audit	Retain according to regulatory/control requirements; export to security evidence destination	Security/audit and restricted platform roles
Tier 0/1 application operations	Regional bucket with incident and troubleshooting retention	Service owner and SRE, purpose-scoped
Debug/verbose	Short retention, sampled or excluded from central storage when safe	Engineering, nonproduction by default
Data-quality and lineage events	Retain with product evidence and governance metadata	Domain, governance and SRE
AI prompts/responses	Do not log raw content by default; use safe hashes, policy decisions and evaluation metadata	Restricted AI/security roles

Regional logging bucket and organization sink example. Validate filter, writer identity, location and retention against the enterprise control mapping.

```
# Purpose: Regional operational log bucket and organization-level sink.
# Security: the sink writer receives only bucketWriter on the destination bucket.
resource "google_logging_project_bucket_config" "regional_ops" {
  project      = var.logging_project_id
  location     = var.region
  bucket_id    = "egcdp-ops-${var.region}"
  retention_days = var.retention_days
  description  = "Regional production operations and audit telemetry"
  locked       = var.lock_retention
}

resource "google_logging_organization_sink" "regional_ops" {
  name          = "egcdp-prod-${var.region}-ops"
  org_id        = var.organization_id
  destination   = "logging.googleapis.com/projects/${var.logging_project_id}/locations/${var.region}/buckets/${google_logging_project_bucket_config.regional_ops.bucket_id}"
  include_children = true
  filter = <<-EOT
    resource.labels.location="${var.region}"
    AND severity>=DEFAULT
    AND NOT jsonPayload.telemetry_class="debug_nonprod"
  EOT
}

resource "google_project_iam_member" "sink_writer" {
  project = var.logging_project_id
  role    = "roles/logging.bucketWriter"
  member  = google_logging_organization_sink.regional_ops.writer_identity
}
```


12. Google Cloud Managed Service for Prometheus

Managed Service for Prometheus is selected for Prometheus-compatible workload metrics, particularly GKE, hybrid and open-source services. Managed collection is preferred where supported. Self-deployed collectors require an approved exception and operational model.

Control	Standard
Metric naming	Prometheus/OpenTelemetry semantic conventions and enterprise prefixes for custom business metrics.
Labels	No direct personal identifiers; bounded service, region, environment, version and owner labels.
Cardinality	Estimated before release and continuously monitored; uncontrolled IDs are prohibited.
Scrape interval	Aligned to detection requirement and cost; faster is not automatically better.
PromQL	Rules are peer-reviewed, tested against absent/noisy data and maintained as code.
Ownership	Platform owns backend and templates; workload team owns target instrumentation and scrape health.

Illustrative multi-window burn-rate PromQL. Production thresholds must be derived from the approved SLO and alert policy.

```
# Purpose: multi-window burn-rate signal for a 99.9% availability SLO.
# Replace metric and labels with the approved service contract.
(
  (
    1 -
    sum(rate(http_server_requests_total{service_name="analytics-api",code!~"5.."}[1h]))
    /
    sum(rate(http_server_requests_total{service_name="analytics-api"}[1h]))
  ) / (1 - 0.999)
) > 2
and
(
  (
    1 -
    sum(rate(http_server_requests_total{service_name="analytics-api",code!~"5.."}[6h]))
    /
    sum(rate(http_server_requests_total{service_name="analytics-api"}[6h]))
  ) / (1 - 0.999)
) > 6
```

13. OpenTelemetry

OpenTelemetry is the enterprise instrumentation and collection standard for new applications, APIs and agent/tool services. It is an open supporting technology rather than a Google Cloud managed service. Regional gateway collectors implement filtering, enrichment, batching, retries and export to managed backends.

Required resource attribute	Example	Purpose
service.name	finance-position-api	Stable service identity
service.version	2026.07.12-3	Release comparison and rollback analysis
deployment.environment	prod	Environment separation
cloud.region	us-central1	Regional diagnosis and residency
egcdp.owner	grp-fin-data-product	Ownership and routing
egcdp.service_tier	tier0	Priority and readiness policy

Prohibited by default: raw prompts, response bodies, authentication tokens, email addresses, user IDs and arbitrary high-cardinality business keys in metric labels or trace attributes.

Regional OpenTelemetry gateway baseline with memory limits, batching, resource enrichment and removal of sensitive attributes.

```
# Purpose: Regional OpenTelemetry gateway collector baseline.
# Validate exporter names and credentials against current Google Cloud documentation.
receivers:
  otlp:
    protocols:
      grpc:
        endpoint: 0.0.0.0:4317
      http:
        endpoint: 0.0.0.0:4318

processors:
  memory_limiter:
    check_interval: 5s
    limit_mib: 1536
    spike_limit_mib: 384
  resource:
    attributes:
      - key: deployment.environment
        value: prod
        action: upsert
      - key: cloud.region
        from_attribute: gcp.cloud_region
        action: insert
  attributes/security_filter:
    actions:
      - key: enduser.id
        action: delete
      - key: http.request.body
        action: delete
      - key: gen_ai.prompt
        action: delete
  batch:
    send_batch_size: 8192
    timeout: 5s

exporters:
  googlecloud:
    project: egcdp-obs-amer-prod
    user_agent: egcdp-otel-gateway/1.0
```

```
googlemanagedprometheus:
  project: egcdp-obs-amer-prod

extensions:
  health_check:
    endpoint: 0.0.0.0:13133

service:
  extensions: [health_check]
  telemetry:
    logs:
      level: info
  pipelines:
    traces:
      receivers: [otlp]
      processors: [memory_limiter, resource, attributes/security_filter, batch]
      exporters: [googlecloud]
    metrics:
      receivers: [otlp]
      processors: [memory_limiter, resource, attributes/security_filter, batch]
      exporters: [googlemanagedprometheus]
    logs:
      receivers: [otlp]
      processors: [memory_limiter, resource, attributes/security_filter, batch]
      exporters: [googlecloud]
```

14. Cloud Trace and Error Reporting

Cloud Trace supports latency and dependency diagnosis for instrumented applications and orchestration flows. Error Reporting groups supported application errors for triage. Neither replaces SLO measurement, logs, incident management or product-level quality controls.

Capability	Use	Guardrail
Distributed traces	Critical API and agent journeys, asynchronous handler correlation and latency analysis	Risk-based sampling; no payload or secret leakage
Trace-log correlation	Trace ID in structured logs and supported telemetry	Correlation ID must not be a personal identifier
Error grouping	New or rapidly recurring software errors	Do not page every exception; route based on user impact
Release comparison	Service/version attributes on traces/errors	Version metadata mandatory for production
Managed data services	Use native metrics/logs where tracing is not exposed	Do not invent unsupported span detail

15. Application Metrics

Signal type	Required examples	Operational use
Traffic	Requests, events, rows, bytes, tokens, active consumers	Demand, capacity and impact scope
Errors	5xx/timeout, invalid contract, failed tasks, rejected writes	Availability and correctness
Latency	p50/p95/p99 request, event and processing latency	User experience and bottlenecks
Saturation	CPU/memory/worker/queue/slot/concurrency utilization	Capacity and imminent risk
Business outcome	Freshness good/total, certification, reconciliation, report completion	Primary service SLI
Dependency	Downstream error, timeout and circuit-breaker state	Root cause and blast radius

Application instrumentation must be asynchronous/non-blocking where practical. Telemetry failure must not cause the business transaction to fail unless the telemetry is itself a mandatory control, such as regulated audit evidence. In that case, the fail-open/fail-closed decision is documented in the service threat model.

16. Pipeline and Data Metrics

Domain	Core signals	Page/ticket behavior
Orchestration	DAG/task success, duration, retries, schedule delay, cutoff risk	Page when Tier 0/1 cutoff or dependency is threatened
Streaming	Input/output rate, system lag, backlog, watermark, DLQ, duplicate/replay indicators	Page on user-impacting latency burn; ticket on capacity trend
Batch/Dataflow/Spark	Job state, worker utilization, skew, failed records, reconciliation	Page critical job failure; ticket recurring inefficiency
Data freshness	Age of latest approved partition/event and source-to-product latency	Primary page-level symptom for data products
Data quality	Critical rule pass, completeness, validity, uniqueness, schema/contract	Block certification/page according to product tier
Lineage/metadata	Lineage event delivery, catalog registration, ownership drift	Ticket unless audit/control gap creates high impact
Consumer usage	Queries, reports, API consumers and last successful access	Impact analysis and deprecation decisions

Do not infer correctness from pipeline success. A green job with stale, incomplete or duplicated business data is a failed service outcome.

17. BigQuery, Dataflow and Pub/Sub Metrics

Service	Required operational signals	Correlation pattern
BigQuery	Job success/error, query latency, queued time, slot utilization, bytes processed, reservation/assignment health, storage growth	Correlate query/job labels with service, product, user persona, reservation and release.
Dataflow	Job state, system lag, data freshness, throughput, worker utilization, autoscaling, errors and custom Beam counters	Correlate with Pub/Sub backlog, source rate, release and downstream writes.
Pub/Sub	Oldest unacked message age, undelivered messages, delivery attempts, publish/ack rates, dead-letter activity and quota errors	Correlate subscription to pipeline, region, data product and consumer SLO.

Service dashboards must distinguish workload errors from service/platform errors. BigQuery invalid SQL and authorization failures are tracked separately from platform availability. Pub/Sub metric gaps are handled through multiple indicators and telemetry canaries. Dataflow custom counters are versioned with the pipeline contract.

Operations Dashboard Hierarchy

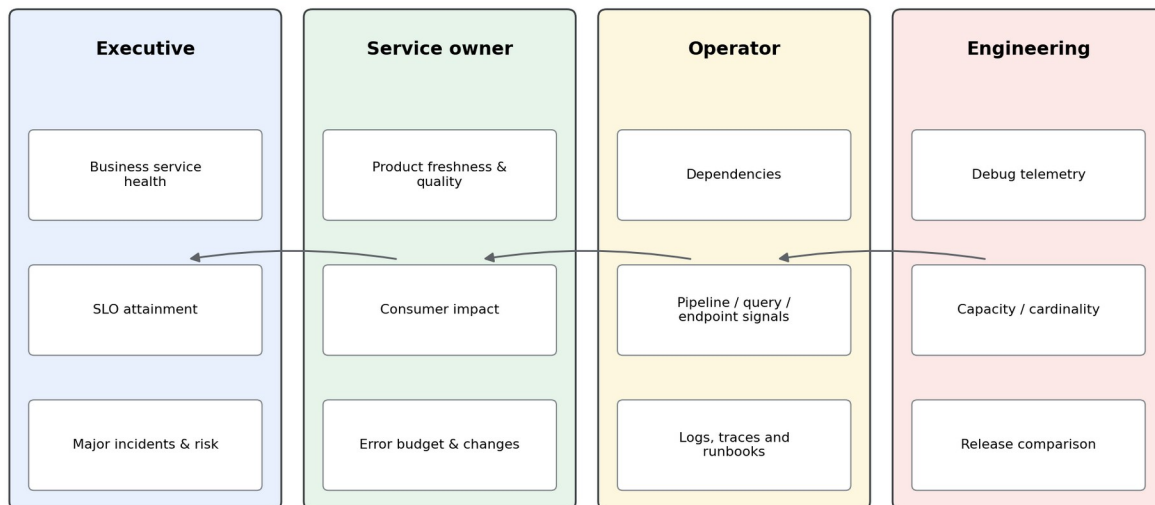


Figure 4 - Dashboard hierarchy from engineering detail to executive service health

18. Vertex AI, Security and FinOps Metrics

Area	Signals	Ownership
Vertex AI online services	Request count, response errors, latency, endpoint/resource utilization, model/version, dependency and quota signals	AI platform/product owner
MLOps/LLMOps	Pipeline success, registry/deployment state, evaluation threshold, drift/feature drift, prompt/model version	AI platform and responsible-AI governance
RAG/agents	Retrieval success, grounding/citation coverage, tool errors, policy/guardrail decisions, token/latency and human-escalation rate	AI product owner and security
Security	Audit delivery, SCC findings, VPC SC denials, key/secret access, sink health, privileged activity and containment actions	Security Operations
FinOps	Spend, forecast, anomaly, reservation/commitment utilization, telemetry cost, allocation completeness and unit economics	FinOps and resource owner

Security and cost signals usually create security incidents or operational tickets, not availability pages. They become page-level when a mandatory control fails, evidence is at risk, or business service impact is imminent.

19. Dashboard Standards and Service Health

- One dashboard has one audience and decision purpose.
- Top row shows user impact, SLO status, incident state and regional scope.
- Component metrics follow the dependency path; raw log panels are links, not substitutes for metrics.
- Every production dashboard shows owner, service ID, environment, region, SLO, runbook and last review date.
- Use consistent units, time zones, percentile labels, missing-data behavior and change/release annotations.
- Executive dashboards show business services and risk, not CPU charts.

Dashboard	Primary Audience	Required Content	Cadence	Design Guardrail
Executive service health	Executives and product leaders	Tier health, SLO attainment, major incidents, risk, business impact	Daily/real time	No raw infrastructure noise
SRE command center	SRE and incident command	Active incidents, burn rates, telemetry health, regional status, dependencies	Real time	Single source during incidents
Data-product health	Domain owners and stewards	Freshness, quality, volume, lineage events, consumer impact	Real time/daily	Per product and region
Pipeline operations	Data engineering	DAG status, Dataflow lag, Pub/Sub backlog, retries, DLQ, reconciliation	Real time	Correlated by pipeline ID
Analytics operations	Analytics platform	BigQuery jobs, queued time, reservation utilization, BI health, query failures	Real time/daily	Separate user errors from platform errors
AI operations	AI platform	Endpoint traffic, latency, errors, model/drift/evaluation signals, token usage	Real time/daily	Segment by model/version/risk tier
Security operations	SOC	Audit, SCC findings, perimeter denials, key access, sink health	Real time	Security-owned access
FinOps operations	FinOps and owners	Spend, forecast, anomaly, unit cost, allocation completeness	Daily/monthly	No paging except control failure

Representative SRE command-center dashboard definition. Production dashboards must be deployed and reviewed as code.

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{
  "displayName": "EGCDP SRE Command Center",
  "mosaicLayout": {
    "columns": 12,
    "tiles": [
      {
        "width": 6,
        "height": 4,
        "widget": {
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          "scorecard": {
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              "prometheusQuery": "max(egcdp_slo_burn_rate{environment=\"prod\"})"
            }
          }
        }
      },
      {
        "xPos": 6,
        "width": 6,
        "height": 4,
        "widget": {
          "title": "Telemetry canary age",
          "scorecard": {
```

```

        "timeSeriesQuery": {
          "timeSeriesFilter": {
            "filter": "metric.type=\"custom.googleapis.com/observability/canary_age_seconds\"",
            "aggregation": {
              "alignmentPeriod": "60s",
              "perSeriesAligner": "ALIGN_MAX"
            }
          }
        }
      },
    ],
    {
      "yPos": 4,
      "width": 12,
      "height": 5,
      "widget": {
        "title": "Open incidents by severity",
        "xyChart": {
          "dataSets": [
            {
              "timeSeriesQuery": {
                "timeSeriesFilter": {
                  "filter": "metric.type=\"custom.googleapis.com/incidents/open\"",
                  "aggregation": {
                    "alignmentPeriod": "300s",
                    "perSeriesAligner": "ALIGN_MAX",
                    "crossSeriesReducer": "REDUCE_SUM",
                    "groupByFields": [
                      "metric.label.severity"
                    ]
                  }
                }
              }
            }
          ],
          "plotType": "STACKED_BAR"
        }
      }
    }
  ],
  "labels": {
    "owner": "platform-sre",
    "environment": "prod"
  }
}

```

[Excerpt shown; full dashboard.json is included in the companion package.]

20. Alerting Standards, Severity and Routing

Severity	Definition	Initial response	Routing
SEV0	Enterprise-wide or catastrophic regulatory/safety/customer impact	Immediate; executive and crisis command	Primary and secondary on-call, incident commander, executives, security/legal as required
SEV1	Major Tier 0/1 service impact or rapid error-budget burn	<=15 minutes	Owning on-call and platform SRE; business owner communication
SEV2	Degraded service with workaround or limited scope	<=1 hour / agreed coverage	Owning team; incident coordination as needed
SEV3	Low-impact defect, trend or operational hygiene	Business-day ticket	Backlog owner

Alert quality gate: every page has an owner, severity, runbook, SLO or documented control objective, tested notification channel, deduplication key, missing-data behavior and closure criteria.

ID	Signal	Severity	Condition	Route	Accountable Owner	Runbook	Response Target	Guardrail
A-001	Telemetry blind spot	Critical	No critical service telemetry for 5 minutes	Page platform SRE	Observability platform	OBS-001	Immediate	Suppress only during approved maintenance
A-002	Tier 0 freshness burn	Critical	2x burn/1h or 6x burn/6h	Page domain on-call + platform SRE	Data product owner	DP-001	Immediate	User-impact aligned
A-003	Pub/Sub backlog growth	High	Oldest unacked age and backlog slope exceed limits	Page streaming on-call	Streaming platform	STR-002	15 min	Correlate with Dataflow lag
A-004	Dataflow job failed	High	Critical production job enters failed state	Page data engineering	Pipeline owner	PIPE-003	15 min	Auto-restart only when safe
A-005	BigQuery reservation saturation	Medium	Sustained queued time and slot utilization	Ticket analytics platform	Analytics platform	BQ-004	4 h	Capacity review if recurring
A-006	Vertex AI endpoint error burn	High	5xx/timeout rate burns SLO	Page AI platform	AI product owner	AI-005	15 min	Include model/version dimension
A-007	Security log sink failure	Critical	Expected sink delivery gap or permission denied	Page security operations	Security platform	SEC-006	Immediate	Preserve evidence and verify no loss
A-008	Budget anomaly	Medium	Spend anomaly exceeds approved threshold	Ticket FinOps and owner	FinOps	FIN-007	1 business day	Not an availability page
A-009	Data-quality rule failure	High	Certified product critical rule fails	Page/ticket based on product tier	Domain owner	DQ-008	15 min	Block certification where mandatory
A-010	Error budget exhausted	High	Remaining budget <= 0	Open change-control action	Service owner	SRE-009	Same day	Freeze risky changes unless exception

21. On-call Model and Escalation

Layer	Coverage	Responsibilities
Primary service on-call	Aligned to service tier	Acknowledge, validate, diagnose and execute service runbook.
Secondary/platform on-call	24x7 for critical shared platforms	Support dependencies, capacity, regional platform and major incident response.
Incident commander	Activated for SEV0/1	Own command, decisions, timeline, roles, communication and safe mitigation.
Business/service owner	On-call/contact path for Tier 0/1	Approve business tradeoffs, consumer communication and risk exception.
Security/privacy/legal	Event-driven escalation	Containment, evidence, notification and regulatory decision.
Vendor support	Contract-aligned	Escalate product/service issue with diagnostics, impact and support entitlement.

- Rotations must have sustainable staffing, documented handover, training and backup coverage.
- On-call engineers require least-privilege standing access and time-bound privileged escalation through approved mechanisms.
- No single expert may be the only path to restore a Tier 0/1 service.
- Alert acknowledgement is not incident resolution; unacknowledged pages automatically escalate.

22. Runbooks and Playbooks

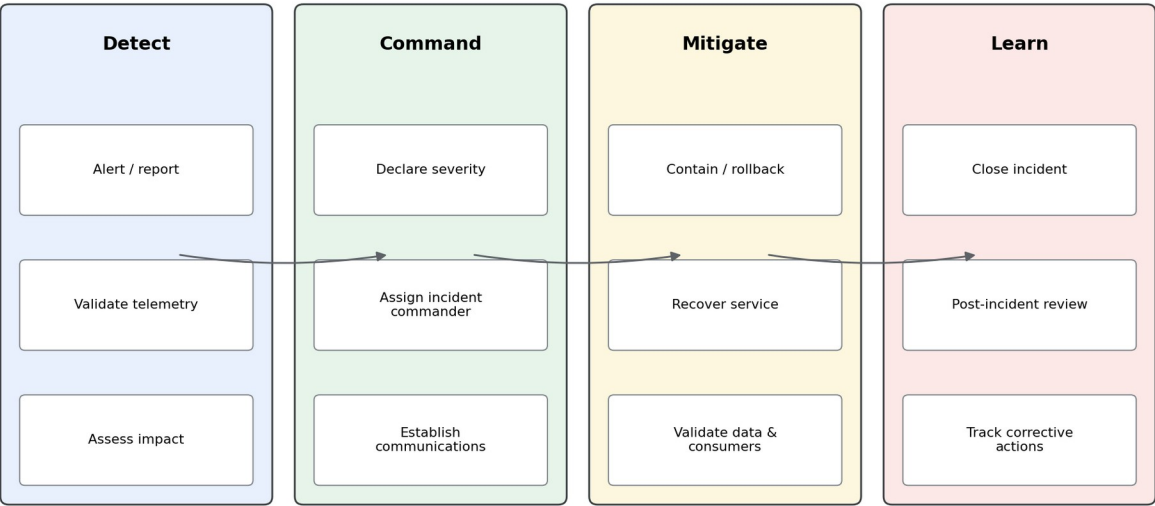
A runbook is service/signal specific and tells an operator how to diagnose and mitigate. A playbook coordinates a cross-service scenario such as regional failure, data corruption, security compromise or major supplier outage. Both are versioned, tested and referenced directly from alerts.

Required runbook section	Control objective
Trigger and user impact	Explain why the alert matters and likely affected consumers.
Safety guardrails	Prevent destructive, irreversible or wide-blast-radius action.
First five minutes	Provide deterministic validation and escalation.
Diagnosis	Link queries, dashboards and dependency checks.
Mitigation	Prioritized reversible actions with preconditions and stop conditions.
Recovery validation	Data reconciliation, quality, freshness, security and consumer confirmation.
Communication	Audience, cadence and regulatory/privacy paths.
Evidence and closure	Timeline, commands, approvals, results and follow-up.

The package contains runbooks/runbook_template.md and STR-002_pubsub_backlog.md.

23. Incident Management

Incident Response Workflow



Legend: solid arrows show primary telemetry, control or accountability flow. Regional and trust boundaries must be validated per workload.

Figure 5 - Incident response workflow

Phase	Key decisions	Required evidence
Detect and validate	Is the signal real? Is telemetry healthy? What is user/business impact?	Alert, corroborating signal, service/SLO and initial scope
Declare and command	Severity, incident commander, roles, bridge and communication cadence	Incident record, role assignments and timeline
Contain and mitigate	Rollback, isolate, scale, fail over or suspend change	Commands, approvals, change/release IDs and safeguards
Recover and validate	Is the service, data and consumer journey genuinely restored?	SLO trend, reconciliation, quality and consumer confirmation
Close and learn	When is monitoring complete? What prevented or delayed recovery?	Closure decision, PIR and action tracker

Machine-readable incident event schema for integration with enterprise ITSM, analytics and audit evidence.

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{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "EGCDP Incident Event",
  "type": "object",
  "required": [
    "incident_id",
    "service_id",
    "severity",
    "state",
    "event_time",
    "owner",
    "summary"
  ],
  "properties": {
    "incident_id": {
      "type": "string",
      "pattern": "^INC-[0-9]{4}-[0-9]{6}$"
    }
  }
}
```

```

},
"service_id": {
  "type": "string"
},
"severity": {
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    "SEV0",
    "SEV1",
    "SEV2",
    "SEV3"
  ]
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  "enum": [
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    "MONITORING",
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  "maxLength": 1000
},
"runbook_id": {
  "type": "string"
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"slo_id": {
  "type": "string"
},
"change_id": {
  "type": "string"
},
"evidence_uri": {
  "type": "string"
}
}
}

```

24. Problem and Change Management

Discipline	Trigger	Required action
Problem management	Repeated incidents, recurring SLO miss, chronic toil, capacity or dependency fault	Create problem record, analyze systemic cause, prioritize permanent corrective action and verify closure.
Standard change	Low-risk, repeatable and pre-approved	Automated evidence, bounded scope and rollback.
Normal change	Material production risk	Peer review, test evidence, SLO/error-budget check, owner approval and scheduled rollback window.
Emergency change	Active severe incident or urgent security exposure	Incident command approval, minimum safe review, complete evidence and mandatory retrospective.
Error-budget policy	Budget exhausted or burn trend unacceptable	Freeze nonessential risky change or obtain accountable exception with expiry and mitigation.

Change records link deployment version, Terraform plan, pipeline/model/prompt version, dashboards, SLO state and incident history. Release annotations are visible on operational dashboards. Volume 9 must implement promotion and rollback workflows that satisfy these interfaces.

25. Capacity and Performance Management

Capability	Capacity indicators	Decision
BigQuery	Queued time, reservation utilization, autoscaling, concurrency, bytes processed and workload class	Adjust assignments/reservations, optimize SQL/storage, isolate workloads or revise demand.
Dataflow	Worker utilization, autoscaling range, system lag, throughput and shuffle/state behavior	Tune workers, pipeline design, streaming mode, quotas or regional capacity.
Pub/Sub	Publish/ack throughput, backlog slope, oldest age, quota errors and subscriber capacity	Scale consumers, tune flow control, adjust quotas or partition workload.
Vertex AI	Endpoint traffic, latency, error, compute utilization, quota and model resource profile	Scale endpoints, adjust model/deployment, quotas, caching or request controls.
Observability	Metric series/cardinality, logs by class, collector saturation, query and alert inventory	Tune retention, sampling, label budget, collectors and scopes.
People/process	Page volume, MTTA/MTTR, toil, coverage gaps and training	Change staffing, automation, support tier and service design.

Capacity planning uses demand forecasts, growth scenarios, business calendars, known campaigns, recovery headroom and quota lead times. Cost is considered jointly with reliability: unused headroom is not automatically waste, but it must be linked to a service objective or recovery need.

26. Operational and Production Readiness

Gate	Minimum evidence	Decision authority
Operational Readiness Review (ORR)	Owner, tier, dependencies, telemetry contract, dashboards, alerts, on-call, runbooks, capacity, support and failure tests	Platform SRE and service owner
Production Readiness Review (PRR)	Security/privacy, data governance, deployment/rollback, SLO, error budget, DR, FinOps, operational evidence and unresolved risks	Architecture/Security/Production review authorities
Material change re-review	Changed dependency, region, model, data classification, SLO, scaling or failure behavior	Relevant service owner and control authority
Exception	Business justification, risk, compensating control, owner, expiry and remediation date	Accountable risk authority

Readiness is evidence-based. A completed checklist without alert tests, runbook exercise, rollback result and measurable SLI is not approval evidence.

27. Post-incident Review and Reliability Improvement

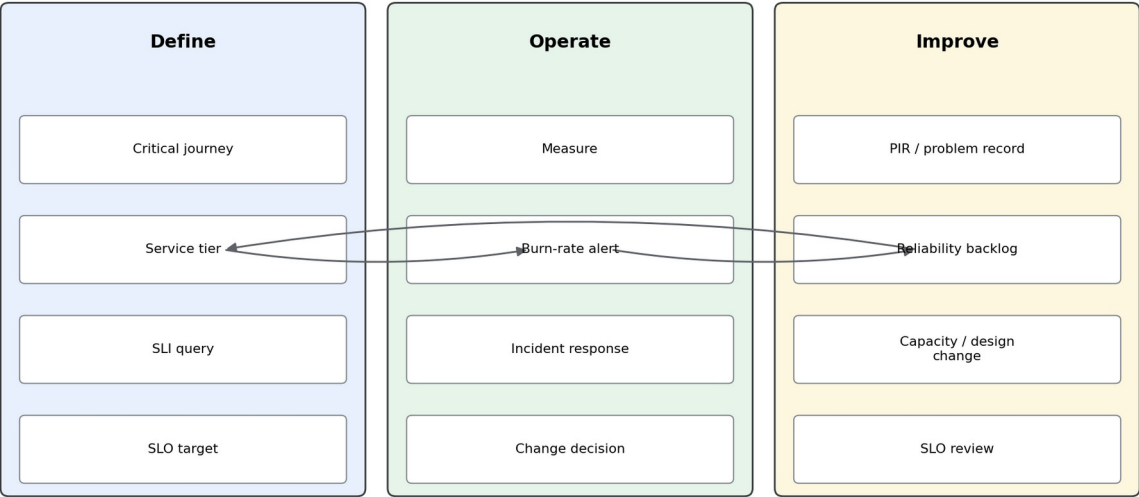
SEV0/1 and material SEV2 incidents require a blameless post-incident review. The objective is to understand system conditions, decisions and organizational factors, not to assign personal fault.

PIR element	Required content
Impact	Business, consumer, data, security, regulatory, region and duration.
Timeline	Detection, declaration, decisions, mitigations, recovery, validation and communication.
Detection analysis	What detected the incident, what should have, and telemetry gaps.
Contributing factors	Technical, process, ownership, capacity, dependency and change factors.
What worked/did not	Runbooks, automation, communication, access, vendor support and recovery.
Corrective actions	Owner, priority, due date, verification and linked problem/change record.
Learning distribution	Reusable pattern, standards update, training and risk-register change.

PIR metrics include detection time, acknowledgement, mitigation, recovery, time to consumer confirmation, number/age of corrective actions, recurrence and toil. MTTR is not interpreted without incident severity and scope.

28. Chaos Engineering and Resilience Testing

SLO and Error-Budget Lifecycle



Legend: solid arrows show primary telemetry, control or accountability flow. Regional and trust boundaries must be validated per workload.

Figure 6 - SLO, error-budget and reliability improvement lifecycle

Test type	Examples	Safety requirement
Telemetry failure	Collector stopped, sink permission removed in nonproduction, notification channel disabled	Canary detects blind spot; no evidence loss in production.
Dependency degradation	Delayed source, quota reduction, downstream timeout, schema incompatibility	Bounded scope, rollback and consumer impact threshold.
Data failure	Late/duplicate records, failed quality rule, corrupt partition in test data	No production restricted data; reconciliation proves detection.
Capacity stress	Backlog, query concurrency, slot/worker saturation and endpoint load	Approved load window and stop conditions.
Regional/service exercise	Loss of regional dependency or control-plane component	Coordinate with Volume 12; no uncontrolled failover.
Human/process exercise	Pager failure, unavailable primary, vendor escalation and executive communication	Observers, explicit objectives and learning review.

Chaos is not random production disruption. Each experiment has a hypothesis, blast radius, owner, abort condition, rollback, data/security controls and measurable result.

29. Service Health, Support and Vendor Escalation

Capability	Enterprise standard
Service health	Combine consumer SLOs, regional dependency status, active incidents and service-provider health information.
Support entitlement	Map Tier 0/1 services to appropriate support plan, named contacts and escalation authority.
Vendor case	Include project/service IDs, timestamps/time zone, region, business impact, reproducible evidence, recent changes and safe diagnostics.
Escalation	Define time-based and impact-based escalation through technical account, support and executive channels.
Data handling	Sanitize attachments and logs; do not send Restricted/Regulated data without approved secure process.
Case closure	Verify resolution, document provider guidance, update runbook/risk/architecture and track recurrence.

Provider service health does not replace internal telemetry. Internal incidents may be caused by configuration, quota, identity, data contract, dependency or release failures even when the managed service is healthy.

30. Structured Service Assessments

The following assessments use the 32-field structure mandated by the master prompt. They are decision records for selected observability services and supporting technology, not generic product catalogs. Current quotas, locations, CMEK, VPC Service Controls, commercial terms and preview/GA status must be validated at implementation.

30.1 App Hub and Application Monitoring - Structured Assessment

Assessment field	Assessment
Service name	App Hub and Application Monitoring
Purpose	Organize application resources and provide application-centric operational context where available.
Enterprise architecture role	Conditional service inventory and application topology layer for business-service operations.
Business capabilities supported	Service ownership, dependency visibility and application-centric monitoring.
Technical capabilities	Application registration, services/workloads and application monitoring capabilities subject to product maturity and location.
Benefits	Connects resource telemetry to application/service context.
Limitations	Capabilities and availability may evolve; not all data/AI managed resources map uniformly.
Service quotas	Validate application/resource registration and API limits.
Regional availability considerations	Validate supported locations and data-residency behavior.
High-availability characteristics	Managed control plane; source-of-truth ownership registry must remain exportable.
Disaster-recovery considerations	Recreate application definitions from code; do not rely on it as sole dependency inventory.
Security model	IAM-scoped application visibility and metadata protection.
IAM requirements	Separate application administrator, editor and viewer roles.
Network requirements	Primarily control-plane; validate APIs and perimeter access.
VPC Service Controls considerations	Validate current support for App Hub/Application Monitoring APIs.
Encryption capabilities	Google-managed encryption.
CMEK support	Validate current support and need.
Logging	Audit administrative changes and registration actions.
Monitoring	Monitor registration drift and missing critical resources.
Alerting	Use underlying Cloud Monitoring policies; application context enriches alerts.
Cost drivers	Service and telemetry features may have product-specific costs; validate current commercial model.
Cost-optimization techniques	Register production-critical applications first; avoid stale duplicate inventories.
Operational responsibilities	Enterprise architecture owns taxonomy; service owners maintain application membership.
Integration points	Cloud Monitoring, asset inventory, service catalog and incident management.
Alternatives	Enterprise CMDB/service catalog or third-party application topology.
Selection criteria	Conditional adoption when supported and useful for business-service context.
Best practices	Treat as synchronized view, not sole source; automate registration and ownership validation.
Reference architecture	Application records connect services/workloads to dashboards, SLOs and incident records.
Sample configuration	Application resource definition maintained through supported API/IaC.
Common pitfalls	Unowned applications, stale membership, assuming full topology coverage.
Anti-patterns	Replacing the enterprise service registry without reconciliation.
Production-readiness checklist	Feature/region support, ownership sync, IAM, drift controls and exit strategy

Assessment field	Assessment
	approved.

30.2 Cloud Logging - Structured Assessment

Assessment field	Assessment
Service name	Cloud Logging
Purpose	Central collection, routing, retention, search and export of platform, audit and application logs.
Enterprise architecture role	Authoritative operational and audit event pipeline with regional storage and downstream integrations.
Business capabilities supported	Troubleshooting, auditability, security detection, incident evidence and compliance retention.
Technical capabilities	Log Router, sinks, buckets, views, exclusions, log-based metrics and analytics.
Benefits	Native Google Cloud integration, centralized routing, configurable retention and access.
Limitations	Routing is not retroactive; exclusions can create evidence gaps; high log volume is costly.
Service quotas	Validate ingestion, sink, bucket, view and query limits for each region and organization.
Regional availability considerations	Use regional log buckets where residency requires; map source, storage and export destinations.
High-availability characteristics	Google-managed; customer must validate sink permissions and downstream availability.
Disaster-recovery considerations	Version sink/bucket definitions; protect exported evidence and test restore/query paths.
Security model	IAM on buckets/views, audit logs, least-privilege sink writers, sensitive-data controls.
IAM requirements	Dedicated writer identities; separate log viewer, private log viewer and configuration roles.
Network requirements	Agents/collectors require approved egress; exports require destination access.
VPC Service Controls considerations	Validate supported logging APIs and destinations across service perimeters.
Encryption capabilities	Encrypted at rest/in transit.
CMEK support	Use CMEK only where supported and justified; test key-loss and rotation scenarios.
Logging	Self-audit administrative changes to sinks, exclusions, buckets and views.
Monitoring	Monitor routed volume, errors, expected-event canaries and destination health.
Alerting	Alert on sink failures, permission denials, missing audit events and routing anomalies.
Cost drivers	Ingestion, retention, query/analytics and downstream export/storage.
Cost-optimization techniques	Exclusion filters, retention tiers, sampling non-audit debug logs, route once, avoid duplication.
Operational responsibilities	Platform owns routing; security owns audit retention; teams own application logging quality.
Integration points	Cloud Monitoring, BigQuery, Pub/Sub, Cloud Storage, SIEM and ticketing.
Alternatives	Third-party log platform or self-managed stack.
Selection criteria	Use for native service logs and centralized enterprise routing.
Best practices	Regional buckets, immutable configuration, source/category retention policy, no secrets in logs.
Reference architecture	Organization/folder sinks route to regional buckets and controlled SIEM/data-lake destinations.
Sample configuration	Terraform <code>google_logging_organization_sink</code> and <code>google_logging_project_bucket_config</code> .
Common pitfalls	Over-broad exclusions, duplicated exports, uncontrolled payloads, missing ownership.

Assessment field	Assessment
Anti-patterns	Treating every log as permanent or sending Restricted data to uncontrolled destinations.
Production-readiness checklist	Routing map, IAM, retention, cost, expected-event tests and incident evidence controls approved.

30.3 Cloud Monitoring - Structured Assessment

Assessment field	Assessment
Service name	Cloud Monitoring
Purpose	Enterprise metrics, dashboards, uptime checks, SLOs and alerting.
Enterprise architecture role	Primary managed observability control plane for Google Cloud services and application telemetry.
Business capabilities supported	Service health, incident detection, SLO governance, executive and operator dashboards.
Technical capabilities	Metrics scopes, dashboards, alert policies, uptime checks, service monitoring, PromQL and integrations.
Benefits	Managed, deeply integrated with Google Cloud, multi-project visibility through scoped architecture.
Limitations	Central scopes require deliberate IAM, quota and residency design; service metrics vary by product.
Service quotas	Validate metrics-scope project limits, alert-policy, dashboard, API and custom-metric quotas before rollout.
Regional availability considerations	Control-plane and telemetry location behavior must be validated against data-residency requirements.
High-availability characteristics	Google-managed service; customer must design resilient notification channels and avoid single scoping-project dependencies.
Disaster-recovery considerations	Store dashboards, alerts and SLO definitions as code; use a secondary operational view for critical services.
Security model	IAM, audit logs, least-privilege viewers/editors, scoped access to sensitive telemetry.
IAM requirements	Separate monitoring viewer, editor, alert-policy author and service-account permissions.
Network requirements	Private collection paths where supported; control egress for agents and collectors.
VPC Service Controls considerations	Validate supported APIs and perimeter patterns; test collection and cross-project access.
Encryption capabilities	Google-managed encryption in transit and at rest.
CMEK support	Validate current CMEK support and scope for telemetry data before relying on it as a requirement.
Logging	Administrative and data-access audit logs where applicable.
Monitoring	Self-monitor alert delivery, metrics ingestion and telemetry canaries.
Alerting	SLO burn rate, absence, threshold, forecast and log-based signals.
Cost drivers	Custom metric volume, API usage, uptime checks and retained/processed telemetry.
Cost-optimization techniques	Prefer native metrics, control cardinality, aggregate early, delete unused dashboards/alerts.
Operational responsibilities	Platform SRE owns foundation; service owners own SLOs and response.
Integration points	Cloud Logging, Managed Prometheus, OpenTelemetry, service metrics, notification systems and SIEM.
Alternatives	Third-party observability suite or self-managed Prometheus/Grafana.
Selection criteria	Use as system of record for Google Cloud operational health and common alerting.
Best practices	Regional scoping projects, ownership metadata, alerts-as-code, tested notification channels.
Reference architecture	Regional observability projects aggregate domain/service telemetry into controlled

Assessment field	Assessment
	enterprise views.
Sample configuration	Terraform google_monitoring_alert_policy with user_labels and runbook link.
Common pitfalls	Orphan alerts, infrastructure-only dashboards, unbounded custom labels.
Anti-patterns	One global project with broad viewer access and no residency/access analysis.
Production-readiness checklist	Scope, IAM, alerts, dashboard ownership, notification tests, cost baseline and canary passed.

30.4 Cloud Trace - Structured Assessment

Assessment field	Assessment
Service name	Cloud Trace
Purpose	Distributed tracing and latency analysis for applications and service interactions.
Enterprise architecture role	Trace plane for user journeys, APIs, event handlers and AI/agent orchestration where supported.
Business capabilities supported	Faster root-cause analysis and dependency/latency optimization.
Technical capabilities	Trace ingestion, span analysis, latency distributions and integrations with OpenTelemetry.
Benefits	Managed tracing integrated with Google Cloud.
Limitations	Sampling can hide rare events; data and AI managed services expose varying trace granularity.
Service quotas	Validate trace ingestion, API and retention limits.
Regional availability considerations	Validate data location and cross-region trace routing requirements.
High-availability characteristics	Managed backend; collectors/SDKs need retry and non-blocking failure behavior.
Disaster-recovery considerations	Tracing is diagnostic, not transactional; store configuration as code and maintain log/metric fallback.
Security model	Do not put secrets, raw payloads or regulated data in span attributes.
IAM requirements	Scoped trace agent/viewer roles.
Network requirements	Collectors/SDKs require approved API connectivity.
VPC Service Controls considerations	Validate trace APIs and exporters within service perimeters.
Encryption capabilities	Encrypted at rest/in transit.
CMEK support	Validate current support before adoption as a requirement.
Logging	Trace SDK/collector errors to Cloud Logging.
Monitoring	Trace ingestion success, sampling rate and exporter failures.
Alerting	Use metrics for paging; traces primarily support diagnosis.
Cost drivers	Trace volume and collector/network overhead.
Cost-optimization techniques	Risk-based sampling and attribute allowlists.
Operational responsibilities	Platform sets standards; teams define critical journeys and span boundaries.
Integration points	OpenTelemetry, Cloud Monitoring, Cloud Logging and Error Reporting.
Alternatives	Third-party APM or self-managed tracing backend.
Selection criteria	Use for latency-sensitive service and agent journeys.
Best practices	Correlate trace ID in logs, consistent service names, bounded attributes and sampling.
Reference architecture	OTel-instrumented services export spans through regional collectors.
Sample configuration	OTel SDK resource attributes and collector trace exporter.
Common pitfalls	Sampling without documented rationale; high-cardinality/span payload leakage.
Anti-patterns	Paging directly on individual traces or tracing every payload field.

Assessment field	Assessment
Production-readiness checklist	Critical journeys, sampling, privacy, correlation and exporter failure tests approved.

30.5 Error Reporting - Structured Assessment

Assessment field	Assessment
Service name	Error Reporting
Purpose	Group and surface application errors from supported logs and error events.
Enterprise architecture role	Developer and operator triage aid for application/runtime failures.
Business capabilities supported	Faster detection and diagnosis of recurring software errors.
Technical capabilities	Error grouping, stack traces, notifications and links to logs.
Benefits	Low-friction integration with Cloud Logging and runtimes.
Limitations	Not a replacement for SLOs, incident management or complete exception telemetry.
Service quotas	Validate event ingestion and API limits.
Regional availability considerations	Validate service availability and telemetry location requirements.
High-availability characteristics	Managed service; ensure errors are also visible through logs/metrics.
Disaster-recovery considerations	No unique recovery dependency; maintain source logs and code versions.
Security model	Stack traces may contain sensitive values; sanitize messages and restrict access.
IAM requirements	Scoped error viewer/editor roles.
Network requirements	Application logging path must be available.
VPC Service Controls considerations	Validate supported access patterns.
Encryption capabilities	Encrypted at rest/in transit.
CMEK support	Depends on underlying logging support; validate current behavior.
Logging	Consumes supported error logs/events.
Monitoring	Track error-event delivery and unexpected grouping changes.
Alerting	Use for actionable new/high-rate error groups, not all exceptions.
Cost drivers	Underlying logging/telemetry volume.
Cost-optimization techniques	Deduplicate, sample noisy expected errors and fix root causes.
Operational responsibilities	Application teams own triage; SRE governs severity and incident linkage.
Integration points	Cloud Logging, issue trackers and source repositories.
Alternatives	Third-party APM/error tracker.
Selection criteria	Use as supplemental application-error view.
Best practices	Structured errors, release/version labels, ownership and runbook links.
Reference architecture	Application errors flow to Cloud Logging and Error Reporting; SLO metrics drive pages.
Sample configuration	Structured log event with service, version, trace and exception.
Common pitfalls	Alerting every error, missing release version, leaking data in stack traces.
Anti-patterns	Treating grouped errors as availability or business-impact SLOs.
Production-readiness checklist	Access, sanitization, ownership, notification and incident-linking tested.

30.6 Google Cloud Managed Service for Prometheus - Structured Assessment

Assessment field	Assessment
Service name	Google Cloud Managed Service for Prometheus
Purpose	Managed collection, storage and query of Prometheus-format metrics.
Enterprise architecture role	Standard Kubernetes and open-source workload metrics plane integrated with Cloud Monitoring.
Business capabilities supported	Application and infrastructure reliability for GKE, hybrid and portable workloads.
Technical capabilities	Managed collectors, self-deployed collectors, PromQL, rule evaluation and multi-project monitoring.
Benefits	Prometheus compatibility without operating a complete metrics backend.
Limitations	Cardinality and scrape design remain customer responsibilities; collection modes require planning.
Service quotas	Validate time-series ingestion, label/cardinality and query limits.
Regional availability considerations	Validate collection and storage location behavior for residency-sensitive workloads.
High-availability characteristics	Managed backend; collectors need multi-zone design and buffering expectations.
Disaster-recovery considerations	Configuration as code; redeploy collectors and rules; maintain critical fallback signals.
Security model	Workload Identity, least-privilege metric writer/viewer and protected endpoints.
IAM requirements	Metric writer for collectors; scoped viewer/query roles for operators.
Network requirements	Collectors need connectivity to scrape targets and Google APIs through approved paths.
VPC Service Controls considerations	Validate telemetry ingestion/query across perimeters and hybrid collectors.
Encryption capabilities	Encrypted in transit and at rest.
CMEK support	Validate current support before making CMEK mandatory.
Logging	Collector and rule-evaluator logs to Cloud Logging.
Monitoring	Monitor scrape health, collector queue, ingestion failures and cardinality.
Alerting	PromQL alerting policies and SLO burn-rate rules.
Cost drivers	Ingested samples/time series and related monitoring usage.
Cost-optimization techniques	Drop unused metrics, control label cardinality, tune scrape intervals, recording rules.
Operational responsibilities	Observability platform owns standard; workload teams own instrumentation and target health.
Integration points	Cloud Monitoring, GKE, OpenTelemetry, Grafana and incident tooling.
Alternatives	Self-managed Prometheus/Thanos or third-party metrics platform.
Selection criteria	Use where Prometheus ecosystem compatibility is required.
Best practices	Managed collection by default, controlled relabeling, label budget and rules-as-code.
Reference architecture	Regional collectors feed managed backend and enterprise metrics scope.
Sample configuration	PodMonitoring/ClusterPodMonitoring plus PromQL alert policy.
Common pitfalls	User IDs in labels, scraping high-cardinality endpoints, no scrape-health alert.
Anti-patterns	Operating duplicate Prometheus backends without a migration or ownership plan.
Production-readiness checklist	Collection mode, identity, network, cardinality, PromQL rules and failure tests approved.

30.7 OpenTelemetry - Structured Assessment

Assessment field	Assessment
Service name	OpenTelemetry
Purpose	Vendor-neutral instrumentation and collection for metrics, logs and traces.
Enterprise architecture role	Enterprise telemetry contract and portability layer for applications and data/AI services.
Business capabilities supported	Consistent observability across Google Cloud, hybrid and multi-cloud workloads.
Technical capabilities	SDKs, semantic conventions, Collector receivers/processors/exporters, sampling and enrichment.
Benefits	Open standard, reusable instrumentation and controlled routing.
Limitations	Collector lifecycle, resource sizing, config testing and semantic governance remain customer-owned.
Service quotas	Downstream backend quotas and collector resource limits must be modeled.
Regional availability considerations	Deploy collectors regionally; prevent unintended cross-border export.
High-availability characteristics	Run collectors redundantly; design queues, retries and backpressure.
Disaster-recovery considerations	Configuration and images stored immutably; redeploy per region; maintain direct critical-path metrics where justified.
Security model	mTLS/TLS, workload identity, secret management, payload filtering and least privilege.
IAM requirements	Export identities limited to telemetry write permissions.
Network requirements	Private endpoints or controlled egress; firewall and DNS must support exporters.
VPC Service Controls considerations	Validate exporters against perimeter restrictions.
Encryption capabilities	TLS for collection/export; backend encryption applies at destination.
CMEK support	Depends on destination backend; collector itself does not provide storage CMEK.
Logging	Collector self-logs with sensitive payload controls.
Monitoring	Queue depth, dropped spans/metrics/logs, exporter errors, memory and CPU.
Alerting	Collector saturation, export failures and drop-rate thresholds.
Cost drivers	Collector compute/network and downstream telemetry volume.
Cost-optimization techniques	Tail/head sampling, batching, attribute filtering, aggregation and compression.
Operational responsibilities	Platform publishes golden collector; teams instrument to contract.
Integration points	Cloud Trace, Cloud Monitoring, Managed Prometheus, Cloud Logging and third-party tools.
Alternatives	Vendor-specific agents and SDKs.
Selection criteria	Use for new application/agent instrumentation and multi-cloud portability.
Best practices	Approved semantic conventions, resource attributes, PII filtering, config validation and canary.
Reference architecture	Regional gateway collectors receive OTLP and export to managed Google Cloud backends.
Sample configuration	Collector YAML with memory limiter, batch, resource processor and Google exporters.
Common pitfalls	No queue persistence, unbounded attributes, secrets in spans/logs.
Anti-patterns	Every team invents its own attributes and collector topology.
Production-readiness checklist	Schema, security, capacity, sampling, failure behavior and downstream routing tested.

31. Architecture Decision Records

ADR	Decision	Status
ADR-1101	Adopt regional observability projects with controlled enterprise aggregation	Accepted - architecture baseline
ADR-1102	Use business-service SLOs and error budgets as the primary reliability control	Accepted - architecture baseline
ADR-1103	Centralize log routing regionally with tiered retention	Accepted - architecture baseline
ADR-1104	Standardize on OpenTelemetry and Managed Service for Prometheus	Accepted - architecture baseline
ADR-1105	Page on user impact and SLO burn rate, not raw resource thresholds	Accepted - architecture baseline
ADR-1106	Manage observability configuration as versioned code	Accepted - architecture baseline
ADR-1107	Adopt federated operations with central platform SRE and product on-call	Accepted - architecture baseline
ADR-1108	Treat telemetry as governed enterprise data	Accepted - architecture baseline
ADR-1109	Use gated and reversible operational automation	Accepted - architecture baseline
ADR-1110	Require evidence-based ORR, PRR and post-incident improvement	Accepted - architecture baseline

Complete ADRs using all mandated fields are included in the `adrs/` directory. They must be reviewed when current product capabilities, locations, regulation, pricing, support model or enterprise tooling changes.

ADR-1101: Adopt regional observability projects with controlled enterprise aggregation

Field	Content
Status	Accepted - architecture baseline
Domain	Observability
Problem	A single global observability plane can conflict with data residency, least privilege and blast-radius requirements.
Alternatives considered	Single global observability project; Independent domain-only monitoring; Regional projects with controlled aggregation
Decision	Use AMER, EMEA and APAC observability projects and logging locations, with carefully scoped enterprise aggregation for approved operational metadata.
Tradeoff	Regional deployment adds governance and duplication, but provides clearer residency, access and failure boundaries.
Implementation implication	Publish versioned standards/templates, onboard Tier 0/1 first, and integrate ownership/SLO/runbook metadata with service management.
Review trigger	Major incident, repeated SLO miss, region or regulation change, product capability change or material cost/support change.

Full record: adrs/ADR-1101_*.md

ADR-1102: Use business-service SLOs and error budgets as the primary reliability control

Field	Content
Status	Accepted - architecture baseline
Domain	SRE governance
Problem	Infrastructure uptime does not adequately represent data freshness, quality, analytics availability or AI service outcomes.
Alternatives considered	Infrastructure-only thresholds; Contractual SLAs only; Business-service SLOs with error budgets
Decision	Define SLOs at business-service and data-product boundaries, backed by component indicators; use error budgets to govern change and reliability work.
Tradeoff	Requires ownership and instrumentation maturity; prevents teams from optimizing components while consumers are failing.
Implementation implication	Publish versioned standards/templates, onboard Tier 0/1 first, and integrate ownership/SLO/runbook metadata with service management.
Review trigger	Major incident, repeated SLO miss, region or regulation change, product capability change or material cost/support change.

Full record: [adrs/ADR-1102_*.md](#)

ADR-1103: Centralize log routing regionally with tiered retention

Field	Content
Status	Accepted - architecture baseline
Domain	Logging
Problem	Uncontrolled project-local logs impede incident response, audit and cost governance.
Alternatives considered	Project-local logging; Single worldwide bucket; Regional centralized routing
Decision	Use organization/folder sinks into regional log buckets and controlled SIEM/archive destinations; apply tiered retention and carefully reviewed exclusions.
Tradeoff	Misconfigured sinks or exclusions can create evidence gaps; telemetry canaries and configuration-as-code mitigate.
Implementation implication	Publish versioned standards/templates, onboard Tier 0/1 first, and integrate ownership/SLO/runbook metadata with service management.
Review trigger	Major incident, repeated SLO miss, region or regulation change, product capability change or material cost/support change.

Full record: [adrs/ADR-1103_*.md](#)

ADR-1104: Standardize on OpenTelemetry and Managed Service for Prometheus

Field	Content
Status	Accepted - architecture baseline
Domain	Telemetry
Problem	Multiple agents, SDKs and metric formats create duplicated cost, inconsistent labels and vendor lock-in.
Alternatives considered	Vendor-specific agents only; Self-managed Prometheus/Thanos; OpenTelemetry plus managed Google Cloud backends
Decision	Use OpenTelemetry semantic conventions and collectors for new applications, and Managed Service for Prometheus where Prometheus compatibility is needed.
Tradeoff	Collector operations and label governance remain customer responsibilities.
Implementation implication	Publish versioned standards/templates, onboard Tier 0/1 first, and integrate ownership/SLO/runbook metadata with service management.
Review trigger	Major incident, repeated SLO miss, region or regulation change, product capability change or material cost/support change.

Full record: [adrs/ADR-1104_*.md](#)

ADR-1105: Page on user impact and SLO burn rate, not raw resource thresholds

Field	Content
Status	Accepted - architecture baseline
Domain	Alerting
Problem	Static infrastructure thresholds generate noise and fail to represent urgency.
Alternatives considered	Threshold-only paging; Manual dashboard review; SLO burn-rate and symptom-based paging
Decision	Use multi-window burn-rate alerts and user-impact signals for paging; route capacity and cost signals to tickets unless immediate impact exists.
Tradeoff	Some services lack direct SLI signals; approved proxy indicators are required.
Implementation implication	Publish versioned standards/templates, onboard Tier 0/1 first, and integrate ownership/SLO/runbook metadata with service management.
Review trigger	Major incident, repeated SLO miss, region or regulation change, product capability change or material cost/support change.

Full record: [adrs/ADR-1105_*.md](#)

ADR-1106: Manage observability configuration as versioned code

Field	Content
Status	Accepted - architecture baseline
Domain	DevSecOps
Problem	Manual dashboards, alerts and sinks are difficult to review, promote, reproduce and recover.
Alternatives considered	Console-only management; Central manual operations team; Infrastructure/configuration as code
Decision	Store alert policies, dashboards, SLOs, log routing, collectors and notification metadata in source control and promote through environments.
Tradeoff	Volume 9 and Volume 13 implementation pipelines remain dependencies.
Implementation implication	Publish versioned standards/templates, onboard Tier 0/1 first, and integrate ownership/SLO/runbook metadata with service management.
Review trigger	Major incident, repeated SLO miss, region or regulation change, product capability change or material cost/support change.

Full record: [adrs/ADR-1106_*.md](#)

ADR-1107: Adopt federated operations with central platform SRE and product on-call

Field	Content
Status	Accepted - architecture baseline
Domain	Operating model
Problem	A central team cannot understand or operate thousands of pipelines and data products alone.
Alternatives considered	Central NOC owns all; Domain-only operations; Federated SRE ownership
Decision	Central SRE owns shared observability and incident standards; platform and domain/product teams own their services, SLOs, runbooks and on-call.
Tradeoff	Requires skills, staffing and explicit escalation boundaries.
Implementation implication	Publish versioned standards/templates, onboard Tier 0/1 first, and integrate ownership/SLO/runbook metadata with service management.
Review trigger	Major incident, repeated SLO miss, region or regulation change, product capability change or material cost/support change.

Full record: [adrs/ADR-1107_*.md](#)

ADR-1108: Treat telemetry as governed enterprise data

Field	Content
Status	Accepted - architecture baseline
Domain	Security and governance
Problem	Logs, traces and metrics can contain identifiers, prompts, queries, schema names and regulated information.
Alternatives considered	Unrestricted telemetry collection; Disable diagnostic detail; Governed telemetry contract
Decision	Classify telemetry, enforce attribute allowlists, regional routing, retention, access controls and sensitive-data prevention.
Tradeoff	Filtering can reduce diagnostic detail; teams must design safe correlation identifiers.
Implementation implication	Publish versioned standards/templates, onboard Tier 0/1 first, and integrate ownership/SLO/runbook metadata with service management.
Review trigger	Major incident, repeated SLO miss, region or regulation change, product capability change or material cost/support change.

Full record: [adrs/ADR-1108_*.md](#)

ADR-1109: Use gated and reversible operational automation

Field	Content
Status	Accepted - architecture baseline
Domain	Automation
Problem	Automatic remediation can reduce MTTR but can amplify faults or destroy evidence.
Alternatives considered	Fully manual response; Unrestricted auto-remediation; Gated reversible automation
Decision	Allow automation only for bounded, tested and reversible actions; destructive or cross-region actions require approval and evidence.
Tradeoff	Human gates may increase response time for some scenarios.
Implementation implication	Publish versioned standards/templates, onboard Tier 0/1 first, and integrate ownership/SLO/runbook metadata with service management.
Review trigger	Major incident, repeated SLO miss, region or regulation change, product capability change or material cost/support change.

Full record: [adrs/ADR-1109_*.md](#)

ADR-1110: Require evidence-based ORR, PRR and post-incident improvement

Field	Content
Status	Accepted - architecture baseline
Domain	Operational assurance
Problem	Checklists without evidence do not prove operability or sustained improvement.
Alternatives considered	Informal peer review; Document-only checklist; Evidence-based operational gates
Decision	Require measurable SLOs, alert tests, runbook exercises, rollback evidence and action tracking before production and after major incidents.
Tradeoff	Adds delivery effort but reduces preventable production failure.
Implementation implication	Publish versioned standards/templates, onboard Tier 0/1 first, and integrate ownership/SLO/runbook metadata with service management.
Review trigger	Major incident, repeated SLO miss, region or regulation change, product capability change or material cost/support change.

Full record: [adrs/ADR-1110_*.md](#)

32. Risks and Mitigations

ID	Risk	Impact	Mitigation	Owner	Status
R-1101	Telemetry blind spot caused by failed agent, sink, quota, IAM or routing	High	Synthetic telemetry canaries, routing health checks, dual-path critical audit logs, tested runbook	Observability Platform	Open
R-1102	Alert fatigue hides true incidents	High	SLO/burn-rate alerts, deduplication, ownership, quarterly alert review, no orphan alerts	SRE Council	Open
R-1103	High-cardinality metrics create cost and performance issues	Medium	Metric allowlist, label budget, cardinality tests, aggregation and sampling	Observability Platform	Open
R-1104	Centralized observability violates residency or least privilege	High	Regional telemetry projects, filtered aggregation, scoped access, documented residency mapping	Security Architecture	Open
R-1105	Domain teams treat platform SRE as sole operator	High	Federated ownership, product on-call, SLO accountability, RACI and readiness gates	CDO/SRE	Open
R-1106	Automation amplifies incident impact	High	Gated reversible automation, blast-radius limits, approval for destructive actions, audit logs	Platform SRE	Open
R-1107	SLOs are infrastructure-only and miss business impact	High	Business-service SLIs at data-product and consumer boundaries	Service Owners	Open
R-1108	Monitoring cost grows without control	Medium	Retention tiers, exclusion filters, metric governance, usage dashboards, FinOps reviews	FinOps	Open
R-1109	PIRs do not lead to improvement	Medium	Action owner, due date, verification, trend reporting, recurring-problem governance	Problem Manager	Open
R-1110	Volume 9 CI/CD controls are not yet implemented	High	Treat observability-as-code gates as implementation dependency and track to Volume 9	DevSecOps	Open

33. Implementation Roadmap

Phase	Name	Indicative Duration	Outcomes
0	Mobilize and baseline	2-4 weeks	Service inventory, ownership, tiering, current telemetry assessment, critical gaps
1	Observability foundation	6-8 weeks	Regional observability projects, metrics scopes, logging buckets/sinks, IAM, retention, cost guardrails
2	SLO and alerting MVP	6-10 weeks	Tier 0/1 SLOs, error budgets, burn-rate alerts, dashboards, on-call and runbooks
3	Data/AI observability scale-out	8-12 weeks	Freshness, quality, lineage, BigQuery, Dataflow, Pub/Sub, Vertex AI operational signals
4	Operational governance	6-8 weeks	ORR/PRR, PIR, problem/change/capacity processes, vendor escalation, service health
5	Resilience and automation	Ongoing	Chaos tests, failure injection, gated remediation, cross-region exercises, optimization

Wave	Priority services/data products	Exit criteria
Pilot	One Tier 0 streaming product, one Tier 1 batch product, BigQuery platform and observability foundation	Measured SLOs, tested alerts, runbooks, on-call, regional routing and cost baseline.
Scale 1	All Tier 0 shared platforms and top regulated products	Portfolio command center, incident/PIR process and readiness evidence.
Scale 2	Remaining Tier 1 products, analytics and AI endpoints	Federated ownership and standard telemetry contracts.
Optimize	Tier 2/3 rationalization and legacy-tool convergence	Reduced duplicate tools/noise, cost/cardinality optimization and resilience maturity.

34. Validation Criteria and Acceptance Tests

ID	Test	Acceptance
VAL-001	Generate synthetic success/failure/freshness signals and prove the SLI query calculates the expected ratio/window.	SLO result matches test data and documentation.
VAL-002	Trigger each page-level alert in a controlled environment.	Correct owner/channel receives page with severity, service, runbook and deduplication.
VAL-003	Stop or isolate a regional OpenTelemetry collector.	Secondary path or alert detects drops before the approved blind-spot threshold.
VAL-004	Break a logging sink writer permission in nonproduction.	Expected-event canary detects the routing failure and runbook identifies IAM cause.
VAL-005	Create Pub/Sub backlog and Dataflow lag under approved load.	Dashboard correlates signals; page occurs only when service SLO is at risk.
VAL-006	Exhaust an error budget in test data.	Change policy, dashboard and exception workflow behave as approved.
VAL-007	Insert prohibited sensitive/high-cardinality attributes into telemetry contract test.	Validation fails without exposing values.
VAL-008	Run SEV1 exercise through incident declaration, communication, mitigation and PIR.	Timeline, roles, evidence and actions satisfy the template.
VAL-009	Verify regional access personas and denied cross-region/log access.	Least privilege and residency controls work as designed.
VAL-010	Restore observability configuration from repository into a clean project.	Dashboards, alerts, sinks and SLO definitions reproduce successfully.
VAL-011	Test primary notification-channel failure.	Fallback channel and escalation operate within response target.
VAL-012	Review telemetry cost and cardinality against approved budget.	Usage is attributable; anomalies and top contributors are actionable.

Post-deployment smoke-test starter for policy inventory, telemetry canary and runbook metadata.

```
#!/usr/bin/env bash
# Purpose: post-deployment observability smoke tests.
set -euo pipefail

: "${PROJECT_ID:?set PROJECT_ID}"
: "${SERVICE_ID:?set SERVICE_ID}"
: "${CANARY_METRIC:=custom.googleapis.com/observability/canary_age_seconds}"

echo "Validating active alert policies..."
gcloud monitoring policies list --project "${PROJECT_ID}" \
  --filter="userLabels.service=${SERVICE_ID}" --format="value(name)" | grep -q .

echo "Validating recent telemetry canary..."
gcloud monitoring time-series list \
  --project "${PROJECT_ID}" \
  --filter="metric.type=\"${CANARY_METRIC}\"" \
  --interval="start-time=$(date -u -d '10 minutes ago' +%FT%TZ),end-time=$(date -u +%FT%TZ)" \
  --format=json | python3 -c 'import json,sys; assert len(json.load(sys.stdin)) > 0'

echo "Validating runbook labels are present..."
gcloud monitoring policies list --project "${PROJECT_ID}" --format=json | \
python3 -c 'import json,sys; p=json.load(sys.stdin); assert all(x.get("userLabels",{}).get("runbook_id") for x in p if x.get("enabled", True))'

echo "Observability smoke tests passed."
```

35. Production Readiness Checklist

ID	Control	Classification	Evidence
PR-001	Business service owner, technical owner and on-call are recorded	Mandatory	Ownership registry
PR-002	Service tier, dependencies, RTO/RPO and critical user journeys are approved	Mandatory	Service record
PR-003	SLIs and SLOs are measurable from production telemetry	Mandatory	SLO configuration and query
PR-004	Burn-rate alerts route to tested notification channels	Mandatory	Alert test evidence
PR-005	Runbooks exist for every page-level alert	Mandatory	Runbook catalog
PR-006	Metrics, logs and traces use approved naming and label budgets	Mandatory	Telemetry review
PR-007	Sensitive data is excluded, redacted or access-controlled in telemetry	Mandatory	Privacy/security review
PR-008	Regional logging, retention and residency are documented	Mandatory	Logging design
PR-009	Dashboards separate user impact from component health	Mandatory	Dashboard review
PR-010	Capacity and cost thresholds are baselined	Mandatory	Capacity model
PR-011	Failure, rollback and recovery paths are tested	Mandatory	Exercise evidence
PR-012	ORR/PRR evidence is approved before production	Mandatory	Signed review
PR-013	PIR process and action tracking are operational	Mandatory	PIR template and tracker
PR-014	Vendor support and escalation routes are documented	Mandatory	Support plan
PR-015	Observability configuration is versioned and promoted through controlled environments	Mandatory	Repository and release evidence

36. Dependencies, Decisions and Next Actions

Dependency	Required interface
Volume 4 Data Engineering	Pipeline IDs, metrics, data-quality/freshness events, replay and runbooks.
Volume 5 Data Mesh	Data-product owner, SLO, consumer impact and certification status.
Volume 6 Analytics	Looker/BigQuery operational experience, dashboard/query SLO and KPI correctness.
Volume 7 AI/GenAI	Endpoint, model, pipeline, RAG/agent, evaluation and safety metrics.
Volume 8 Security	Audit, SCC, identity/perimeter/key activity, incident and telemetry data protection.
Volume 9 DevSecOps - pending	Observability-as-code validation, promotion, release annotation and rollback evidence.
Volume 10 FinOps	Telemetry cost, capacity economics, anomaly and unit cost.
Volume 12 DR	Recovery SLO/RTO/RPO, regional exercises and incident/recovery orchestration.
Volume 13 Terraform	Golden modules for monitoring, logging, collectors, IAM and dashboards.

Prioritized next actions:

1. Approve the federated SRE operating model and service-owner accountability.
2. Select pilot Tier 0/1 services and baseline their critical user journeys and telemetry gaps.
3. Create AMER, EMEA and APAC observability project designs and validate location, IAM, perimeter and quota constraints.
4. Publish telemetry contract, SLI/SLO template, alert severity model, runbook and incident/PIR templates.
5. Implement observability-as-code pipelines as part of the outstanding Volume 9 and Volume 13 work.
6. Execute controlled alert, telemetry-failure and incident-command exercises before production approval.
7. Establish monthly service reliability review and quarterly observability cost/cardinality review.

37. References and Companion Assets

Official Google Cloud references to validate during implementation:

- Google Cloud Monitoring documentation, metrics scopes, alerting and service monitoring.
- Cloud Logging routing, sinks, buckets, retention, exclusions and log-based metrics.
- Google Cloud Managed Service for Prometheus managed collection, multi-project monitoring and PromQL.
- OpenTelemetry Collector and Google Cloud exporters/instrumentation guidance.
- Cloud Trace and Error Reporting documentation.
- BigQuery monitoring and INFORMATION_SCHEMA operational views.
- Dataflow monitoring, job metrics and user counters.
- Pub/Sub monitoring metrics and operational guidance.
- Vertex AI monitoring for endpoints, pipelines, model monitoring and operations.
- Google Cloud Architecture Framework - operational excellence, reliability, security and cost optimization.
- Google SRE guidance for service-level objectives, error budgets, incident response and postmortems.

Companion package contents:

- Six diagrams in PNG, SVG, Mermaid, PlantUML, Graphviz DOT and editable draw.io formats.
- Nine implementation starters for OpenTelemetry, Monitoring, Logging, SLOs, PromQL, dashboards, incident schema and validation.
- Seven 32-field service assessments, ten complete ADRs, eight matrices, runbook templates, manifest and checksums.

Current product terminology and capabilities can change. Before implementation, validate official documentation, release stage, regional availability, quotas, IAM, VPC Service Controls, CMEK, retention, pricing and support commitments.